



Play With Your Partitions

This article explains some free and open source tools like GParted, g4u and partimage, which are used for partitioning, disk cloning, partition backup/restore, etc.

Let's face it—hard disk-related activities can never be simple, especially when that disk has data on it. In this article, we'll take a look at some of the tools that can make these activities easier.

A partitioning tool

There have been many operating systems and live CDs that provide partitioning utilities. Often, they turn out to be either proprietary or not able to support all filesystems used for Linux, Solaris, etc. But GParted is one such important partitioning utility that supports many filesystems and comes with a friendly user interface.

GParted (Figure 1) stands for 'GNOME partition editor'. With it, you can create, modify and delete partitions. Although you can definitely install it on any distribution and run it, I wouldn't recommend running it on a booted OS as problems might occur with certain mounted partitions. A better solution is to run GParted from a live CD. GParted itself comes on a separate live CD; boot it and double click on the GParted icon from the UI presented. It is also a part of a live CD called System Rescue CD.

To run GParted from here, first start X by issuing the `startx` command, and then open the GParted application from the X terminal. It can also be run from a bootable USB drive or via a PXE boot.

Once GParted is run, you can use it for the following operations:

- To create a new partition table (by erasing the old one, if one exists). Use this option very carefully. Unlike other operations, **Undo** is not supported for this.
- A new partition can be created from unallocated/free space. Decide on new partitions as primary, extended or logical, as required.
- To delete an existing partition.
- To resize/move existing partitions (Figure 2). Sizes can be increased only when free space is available on any adjacent side.
- To change flags/labels of a partition.
- To format a partition to any supported filesystem—ext3, reiserfs, etc.
- To check filesystems for errors.
- To view partition information.

All requested operations are added to the pending queue, which is performed in sequence upon clicking the **Apply** option.

The following filesystems are supported